Arithmetic-Fair Comparisons of Exact FIR Software: A Cortex-M0 Model Case Study

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***Abstract*—Arithmetic-range information can change an exact FIR implementation decision. On a pinned Cortex-M0 model, replacing wide direct accumulation by two certified int32 partial sums overturns two of four original demand-Hcub transposition advantages. Hcub then costs 4.33%/5.00% more with GCC10 but 24.06%/5.23% less with GCC15, for blocks 1/64. A prospectively frozen synthetic panel reproduces this comparator boundary in 24 of 32 complete widening-required fixed-compiler cells. Eight unavailable native-fit library identities leave the full panel incomplete. Incremental sharing has a different outcome: Hcub loses to ordinary transposition in all 32 fresh wide cells, despite beating old direct/library controls. Exact saved-event analysis separates arithmetic gains from RAM penalties; only original GCC15/block1 retains Hcub's nominal advantages over both stronger controls throughout an arbitrary RAM-weight stress over [0.5,2]. A matched direct-kernel intervention exposes literal one-output knowledge, removes reviewed next-output preparation and lowers GCC15 modeled cost by 31.44%. The direct compiler gap contracts from +70,469.699925 to −11,857.443121 nJ, reversing that specialized pair's ranking. These results support a bounded empirical case study of comparator fairness, representation and compiler knowledge using established techniques. Costs are configuration-specific event estimates; no new FIR algorithm, general selector or physical-energy advantage is established.**

I. Introduction

Giving direct accumulation the same exact-range information as its transposed competitor changes the implementation decision. For the frozen nonsymmetric `speech_minphase48` filter, demand-Hcub transposition had lower modeled CPU energy than both wide specialized direct and unrolled CMSIS in all four compiler/block cells. Retaining the direct history layout but accumulating two certified native-width partials overturns both GCC10 decisions; Hcub retains both GCC15 cells. Arithmetic legality alone cannot predict that split.

Transposition exposes products of one sample with many constants, enabling shared shift/add expressions. It also changes state meanings, widths and lifetimes. We therefore distinguish three contrasts: partial versus wide direct measures width correction within one organization; Hcub versus ordinary transposition measures incremental sharing with suffix equations fixed; and cross-compiler direct binaries, followed by a single block-knowledge intervention, test a generated-code explanation. Hcub versus corrected direct is the implementation choice, combining sharing and representation.

The original strict four-cell reporting rule, H_R, is retrospective (Section 4.2). A separately frozen prospective panel asks whether strengthening the nontransposed comparator overturns an old-control Hcub advantage on fresh widening-required designs, across paired amplitude/stream regimes. The matched direct intervention asks whether exposing a one-output bound inside the callee removes identified next-output preparation and reduces GCC15 cost and the compiler gap.

Range analysis, native partials, transposition, Hcub and specialization are established techniques. Our contribution is finite empirical evidence about which conclusions survive these controls, with exact event sensitivity and portable raw evidence. Field priority and a general implementation selector remain unestablished.

The fresh panel supplies wide-design reversal witnesses while rejecting an extension of beneficial incremental sharing to those arrays. The matched intervention removes the identified preparation, reduces GCC15 cost and reverses the direct compiler ranking. Together they sharpen the original comparison into a boundary account, retaining negative and unavailable outcomes.

II. Closest prior work

De Coster et al. [1] transform graphs for compiled bit-true DSP simulation; their FIR example uses magnitude information to avoid unnecessary overflow handling. Ménard and Sentieys [6] select software word lengths/instructions under an SQNR constraint, including arithmetic and transfer costs. Our full-int16 exact-output contract differs from their accuracy constraint, but width selection and narrow arithmetic are established techniques.

Gačić et al. [2] generate FIR software with SPIRAL and show that blocking and computation order alter runtime at equal arithmetic count. D'Alberto et al. [3] generate fixed-point XScale transforms and optimize runtime and physically measured energy. FIR generation, locality-aware search and energy-oriented selection therefore precede this synthetic boundary comparison. Their physical measurements do not validate our model costs.

Voronenko and Püschel's [4] Hcub shares arithmetic graphs across constant products, with software concerns including register pressure beyond addition count. Its objective is not an executable energy model. Nikov et al. [5] supply our configuration-specific event weights; affine reweighting is elementary sensitivity analysis. GCC documents constant propagation, cloning and interprocedural optimization [7]; fixed-bound specialization is conventional and those documents can-

not identify a historical responsible pass. Clean-copy artifact checks likewise follow established practice [10].

This targeted source challenge is not an exhaustive priority search. Ménard–Sentieys §§4–5.4, SPIRAL FIR §§4–5, XScale and Nikov full texts support the comparisons. Fresh Hcub full-text access supports sharing (§1.2) and the register-pressure caveat (p. 6). De Coster overlap uses fresh indexed primary excerpts plus recorded earlier inspection; unsuccessful recent full-PDF retrieval licenses no exclusion of unseen capabilities.

III. Exact contract and controlled representations

A. Numeric semantics

All filters have 48 frozen signed-int16 coefficients, ordered so that $h[0]$ multiplies the newest sample. Every candidate must accept all $x \in [-32768, 32767]$, including -32768, regardless of the amplitude used for performance measurement. Initial history is zero. For requested samples only,

$$\begin{aligned} S[n] &= \sum_{k=0}^{47} h[k]x[n-k], \\ Q(s) &= \min\{32767, \max\{-32768, \lfloor s/32768 \rfloor\}\}, \\ y[n] &= Q(S[n]), \qquad x[n] = 0 \text{ for } n < 0. \end{aligned}$$

Exactly one final quantization and saturation is permitted. There is no intermediate truncation, saturation or appended convolution tail. In particular, $Q(-1) = -1$; truncation toward zero at that boundary changes the contract. The generated quantizer clamps before narrowing and implements negative floor using C division without assuming arithmetic signed right shift.

For each coefficient, exact independent-input product endpoints are $l_k = \min(-32768h_k, 32767h_k)$ and $u_k = \max(-32768h_k, 32767h_k)$. Summing those endpoints gives tight intervals for the full sum and any specified group. A sign-aligned legal sample window attains each endpoint. A sufficient group-prefix certificate also uses $32768 \sum_{k\in G} |h_k| \le$ INT32_MAX: every product and accumulated prefix within G fits int32. Full intervals, prefix bounds, transposed suffix bounds and Hcub primitive/restored-product identities are separate certificates. C evaluation widths must respect them; widening after an overflowing addition is too late.

B. The original arithmetic correction

The corrected original direct kernel keeps the same chronological history and literal coefficient order. Fixed groups $[0, 2)$ and $[2, 48)$ have coefficient L1 sums 12,778 and 59,731, hence absolute prefix bounds 418,709,504 and 1,957,265,408. Both fit int32. The full L1 sum is 72,509, with absolute bound 2,375,974,912; legal sign-aligned windows attain 2,375,922,255 and −2,375,955,060, establishing that full-sum int32 overflow is possible. Each completed partial is converted separately before the int64 reduction:

```
int64_t exact_sum = (int64_t)partial0 + (int64_t)partial1;
y[j] = quant(exact_sum);
```

The split was fixed for the missing-control experiment. It is neither an optimized partition nor a claim that the direct realization is optimal.

Ordinary transposition computes the current sum from $h[0]x + z_0^{old}$ and updates z_i from $h[i+1]x + z_{i+1}^{old}$, ending with the last product. Ascending updates preserve the required old states. Suffix bounds justify each state's width. Hcub adds independently checked shared expressions and demand-driven emission near their consuming stage. Safe exact operations replace problematic signed shifts.

Figure 1 contrasts the value meanings and lifetimes. Direct state retains input history; transposed state retains weighted suffix sums. Correctness therefore reconstructs the intended values for each representation instead of comparing unlike state bytes. Hcub versus ordinary transposition holds those equations and layout fixed. Hcub versus direct also changes representation; Hcub versus CMSIS changes additional implementation choices. The entire library-relative saving cannot be assigned to sharing.

IV. Measurement, chronology and prospective questions

A. Target and event models

The target is Cortex-M0 Thumb at `-O2`, separate compilation and no LTO. GCC10 reports 10.3.1 20210621; GCC15 is Arm GNU Toolchain 15.2.Rel1, reporting GCC 15.2.1 20251203. The simulator is pinned to commit `c55ea1b192 6b792d7aabb7e3495b55a7b69ef134`. Code and coefficients reside in Flash, with state/input/output in SRAM. Measurement includes dispatch, recurring calls, copies, convolution, quantization, output stores and retained history/state movement, including a final short call. Initialization, checking and printing lie outside. The audited decoder removes the closing marker instruction consistently.

Let N_I include all instructions, N_M native multiplication instructions, B taken branches, R and W SRAM reads/writes, F Flash data reads and N_{BL} branch-with-link instructions. The zero-wait, single-cycle-MUL Arm-table cycle estimate is

$$T = N_I + R + W + F + 2B + N_{BL}.$$

Nikov et al.'s [5] 20 MHz, prefetch-off, zero-wait weights give modeled CPU energy in nJ:

$$\begin{aligned} E = {} & 0.964258(N_I - N_M) + 1.652455N_M \\ & + 2.091986B + 1.109833R + 0.650563W + 0.633621F. \end{aligned}$$

Their reported configuration MAPE is 2.80%, an aggregate model-validation statistic rather than an uncertainty bound for these pairwise margins. The fitted weights may absorb correlation/residual effects; they are not independently measured physical event prices. We retain integer events and exact six-decimal products. The 0.000001 nJ tolerance for historical floating serialization is solely a data-admission tolerance.

TABLE I
THE FIVE PRINCIPAL ORIGINAL FORMS; ALL SATISFY THE SAME FINAL-OUTPUT CONTRACT.

Short name	Original artifact identifier	Controlled role
CMSIS	`cmsis64_unroll`	Historical unrolled wide library control
Wide direct	`spec_direct`	Specialized chronological direct, wide sum
Partial direct	`partial_direct32x2`	Same direct history, two native partials
Ordinary transposed	`trans_const`	Bounded suffix states, ordinary products
Hcub	`trans_hcub_demand`	Same suffix equations, shared constant products

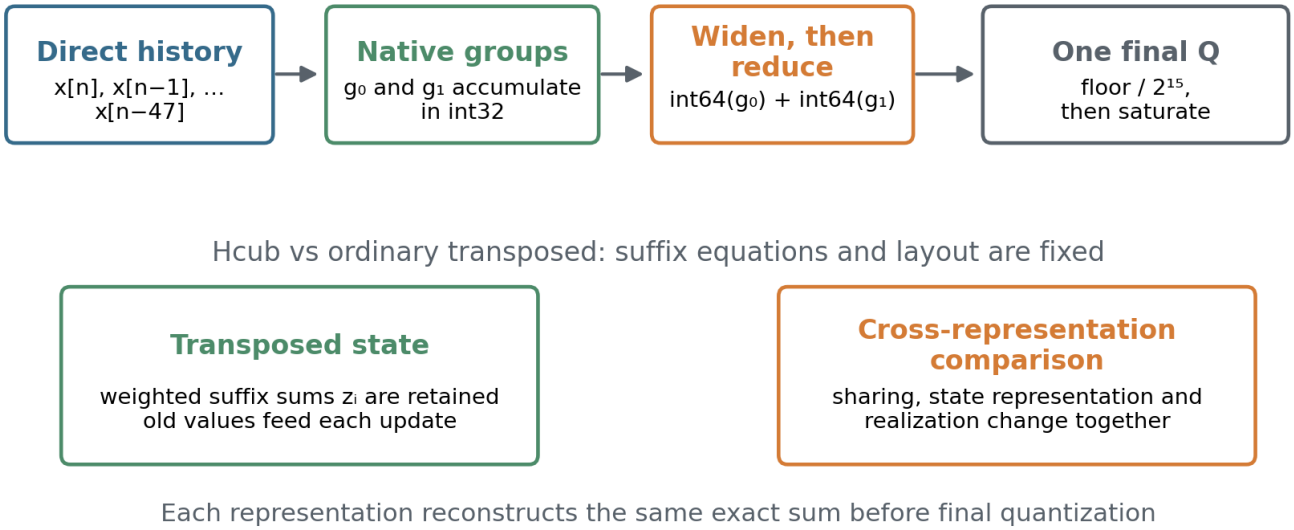


Fig. 1. Exact representations and controlled contrasts. Each direct native group widens before reduction; both representations keep the exact sum until one final quantizer. Direct history and weighted suffix states have different meanings and lifetimes. The cross-representation comparison combines sharing and realization.

B. Admission and evidence chronology

The original pre-evaluation protocol covered three designs, two input guarantees, two blocks and two compilers. A source-level state-traffic estimate was later withdrawn as an optimistic floor: grouped transfers/register retention could reduce it, and no candidate had been excluded using it. Observed unequal width treatment then motivated the fixed two-group/four-cell correction, followed by the compiler account. H_R was formulated afterward, so it is retrospective. Previously inspected filters remain development evidence; the fresh panel and intervention were frozen before target costs.

For the original 257-sample zero-history saved `adc_raw` stream, block1 makes 257 calls; block64 makes four full calls and one one-sample call. The reporting rule is $r = 100(E_H/E_P - 1)$ with Hcub H and partial direct P. H_R survives only if all four admitted values are strictly negative. A tie fails strict superiority; missing/inadmissible evidence makes assessment unavailable. This is a deterministic criterion, without sampling variance or significance testing.

Arithmetic certificates establish full-domain source legality; finite UBSan, target output/state/guard checks address implementation errors. The original independent check admitted 1,028 new, 6,168 historical and 722 signed-edge output comparisons. Emitted-code review confirmed native partials/wide final reduction in all four corrected binaries. Source/binary hashes bind events to checked code. Pinned library semantics, including fast-CMSIS eligibility, require source/toolchain admission beyond finite agreement.

C. Saved-event analysis

For each compiler/block we analyze exactly three oriented pairs: partial minus wide direct, Hcub minus ordinary transposition and Hcub minus partial direct. All use the same 257 outputs. With $v = (N_I - N_M, N_M, B, R, W, F)$ and the weights above, $\Delta E = w \cdot \Delta v$; negative favors the left form. Six-event dominance requires all coordinates nonpositive with at least one strictly negative; mixed signs preclude it. Resources are excluded.

One predeclared arbitrary stress multiplies both RAM weights by λ, leaving the other four fixed. For $\lambda \geq 0$,

$$\Delta E(\lambda) = A + B_{RAM}\lambda,$$
$$B_{RAM} = 1.109833\Delta R + 0.650563\Delta W.$$

The selected closed region is $[0.5, 2]$. It is an uncalibrated coefficient stress, not a confidence interval, hardware configuration or physical-price range. Integer-weight products retain exact decimal values; boundaries are exact rational $-A/B_{RAM}$ values. Rounded roots are display only. Endpoint signs decide strict ordering throughout the region by linearity; any tie defeats strictness. The artifact retains all 12 vectors, cost components, resource deltas and root classifications, including negative algebraic roots.

D. Frozen synthetic panel

Three fresh nonsymmetric gain-normalized 48-tap designs deliberately span one native-sum and two widening-required regimes. With NumPy 1.26.4/SciPy 1.15.3, we apply `firwin(95, cutoff, fs=16000, window=window, scale=True)`, homomorphic `minimum_phase(half=True,n_fft=32768)`, target-L1 scaling and `np.rint` integer rounding. The half-length conversion changes prototype magnitude [8]. These are synthetic arithmetic controls, not representative application filters.

The first cut is the smallest legal two-group boundary; the balanced cut minimizes the larger group L1, breaking ties by smaller boundary. Sign-aligned legal windows attain both full-sum endpoints, proving actual int32 overflow possibility in wide designs. Arrays, certificates, graphs, stimuli and case list were frozen by hash before costs; no outcome-driven replacement occurs.

Each design has six forms: wide direct, first certified direct, ordinary transposed, demand-Hcub and standard unrolled CMSIS, plus native-only fast CMSIS or wide-only balanced direct. All receive identical coefficients, full-int16 semantics, block knowledge and generation budget. Pinned CMSIS-DSP v1.14.4 paths receive reversed coefficients/required storage, with separate source/compiler eligibility and no fast-path scaling [9]. Each Hcub graph uses one unchanged `acm -b 15 -seed 1 -nofs -gc` run on sorted odd fundamentals.

The frozen 1089-sample full-range LCG master and samplewise `full//32` ADC partner are specified in the linked protocol. Startup measures its first 257 outputs from zero history. Post-startup processes 64 samples outside the region, then measures offsets [64,1089), retaining history: 1025 outputs. Both block sizes align at the warm boundary; block64 ends with one short call in each regime. Checks cover the entire prefix, including warmup. The region contrast also changes segment and length, so it does not isolate startup cost.

The cap is 288 performance identities (3 designs × 6 forms × 2 compilers × 2 blocks × 2 amplitudes × 2 regimes). Separate correctness admission uses at most 72 target identities on the frozen 194-sample zero/extremal/edge stream and host UBSan mixed partitions. Actual image/call-chain bounds must fit the original 64 KiB Flash/8 KiB RAM limits. Unsupported and failed cases remain visible; deterministic executions are not independent statistical samples.

For each fixed-compiler cell, define D_{old} as the minimum eligible old nontransposed control (wide direct, standard CMSIS, and fast CMSIS where eligible). D_{new} is the minimum after adding both available certified direct forms. All ties retain every minimizer. The primary extension requires at least one admitted cell of either widening-required design with $E_H < D_{old}$ but $E_H \geq D_{new}$. If no such cell exists among the complete wide-required cells, the extension is refuted; an unavailable required cell leaves the full-panel assessment incomplete. Additional reports retain all-form winners, each certified/wide percentage, Hcub/ordinary percentages, events/resources and all amplitude/regime contrasts. Compiler-selectable decisions minimize each form across compilers first, separately from fixed-compiler results. Reference sums also count the two quantizer clamp conditions, $S \geq 1073741824$ and $S \leq -1073741824$, in each measured region. Output endpoints alone cannot identify those paths.

E. Matched block-knowledge intervention

The intervention fixes the original block1 fixture and 257-sample stream, with two compilers × general/specialized kernels. General kernel bytes, wrapper, ABI and flags remain original. Only expression uses of runtime `b` inside specialized `probe_process` become literal `1`, adding `(void)b;`. Every call meets $b = 1$; arithmetic, groups, quantizer, state/history algorithm and separate compilation/no LTO stay fixed. Four signed-edge target checks and repeated-one-sample host checks support correctness. Failure to reproduce general-control binary bytes/events stops historical causal interpretation.

Define $G_g = E_{15,general} - E_{10,general}$ and $G_s = E_{15,specialized} - E_{10,specialized}$, with saved $G_g = 70469.699925$ nJ. The stronger outcome requires preparation removal in reviewed emitted code, lower GCC15 cost and $G_s < G_g$. Full event/cycle and separate resource deltas test each condition. All total changes belong to block-knowledge specialization; transfer-subset attribution needs additional accounting. The direct-only intervention licenses no other-form ranking or compiler-pass claim.

V. Admitted original comparison and exact saved-event results

A. Width correction changes two decisions

The stronger direct control lowers modeled CPU energy in all four cells, by 32.823801%, 41.860155%, 23.440506% and 41.980694% relative to wide direct, in GCC10/1, GCC10/64, GCC15/1, GCC15/64 order. It weakly improves every one of the six event coordinates in each cell, with strict improvements: the correction is six-event-dominant within this saved comparison. This does not establish physical or resource dominance. All five forms' complete costs remain visible below.

For GCC10/block64, partial direct executes 129,481 instructions and 2,827 MULs; Hcub executes only 116,139 instructions and 257 MULs. Yet Hcub incurs 10,994 additional RAM reads and 18,235 additional writes. Its complete modeled cost is 604.62 versus 575.81 nJ/output. Graph nodes therefore do not determine the implementation winner.

TABLE II

FROZEN ARITHMETIC REGIMES, INDEPENDENTLY PREPARED BEFORE TARGET COSTS. CUTS GIVE FIRST-LEGAL / RANGE-BALANCED TWO-GROUP BOUNDARIES.

Design	Cutoff/window	Target / actual L1	Exact full-sum interval	Direct cuts
`native_lp1800`	1800 Hz / Hamming	50000 / 50001	[−1638421124,1638394411]	Whole native sum
`wide_lp2800`	2800 Hz / Hamming	100000 / 100002	[−3276839180,3276791890]	4 / 5
`wide_lp4600`	4600 Hz / Blackman	110000 / 110003	[−3604545833,3604500772]	3 / 4

TABLE III

ORIGINAL MODELED CPU ENERGY, NJ/REQUESTED OUTPUT; ALL PROCESS THE SAME 257 SAMPLES. DISPLAY ROUNDING DOES NOT DETERMINE WINNERS.

Variant	GCC10 b1	GCC10 b64	GCC15 b1	GCC15 b64
CMSIS	1537.21	905.94	1503.18	838.85
Wide direct	1543.04	990.40	1712.07	1023.10
Partial direct	**1036.55**	**575.81**	1310.75	593.59
Ordinary transposed	1045.78	605.66	1039.99	602.87
Hcub	1081.44	604.62	**995.43**	**562.55**

TABLE IV

$100(E_H/E_{named} - 1)$. BOTH HISTORICAL COMPARISONS FAVOR HCUB IN ALL FOUR CELLS. THE CORRECTED COMPARISON REJECTS H_R WITH TWO SURVIVING CELLS.

Compiler/block	Hcub / wide direct	Hcub / CMSIS	Hcub / partial direct
GCC10/1	−29.91%	−29.65%	+4.330785%
GCC10/64	−38.95%	−33.26%	+5.002907%
GCC15/1	−41.86%	−33.78%	−24.056759%
GCC15/64	−45.01%	−32.94%	−5.229188%

Fixed versus selectable compiler decisions also differ. With GCC15 fixed, Hcub is 24.06% and 5.23% below partial direct. If either compiler may be chosen separately for each form, GCC15 supplies the best tested Hcub and GCC10 the best partial direct; the savings become 3.97% and 2.30%. These minima cover only the tested builds and are not calibrated physical-energy margins.

B. Incremental sharing is smaller and event-mixed

Within the same transposed equations, Hcub costs more at GCC10/block1 and less in the other three nominal cells. Every incremental-sharing event vector is mixed: fewer native MULs and Flash reads coexist with additional RAM transfers. GCC10/block1 has no non-MUL instruction reduction at all.

For GCC15/block64, the 32.94% Hcub/CMSIS saving becomes 6.686823% against ordinary transposition and 5.229188% against corrected direct. Only the first of those stronger contrasts holds suffix equations fixed.

Resource deltas are separate from event dominance. Hcub and ordinary transposition allocate equal state; Hcub's reported process frame is 72 bytes larger in every cell, while selected kernel/coefficient symbols are 36 bytes smaller with GCC10 and 180 with GCC15. Partial and wide direct allocate equal state; partial direct has reported frames smaller by 368, 368, 368 and 376 bytes and selected symbols smaller by 700, 524, 500 and 572 bytes. State allocation, compiler process frame and selected-symbol bytes measure different things and are not added as memory use. Appendix A retains every original principal row.

C. Nominal winners under the arbitrary RAM-weight stress

The exact affine analysis reconciles all 12 nominal differences with the admitted original energy totals. Partial minus wide direct has $A < 0$ and $B_{RAM} < 0$ in every cell, so its six-event result also yields lower modeled cost throughout the selected region and indeed for every $\lambda \geq 0$. All eight Hcub contrasts are mixed, with $A < 0$ and $B_{RAM} > 0$: their non-RAM benefit is eventually outweighed along this chosen stress path.

Figure 2 shows why only GCC15/block1 stays below both controls throughout [0.5,2]. GCC10/block64's incremental benefit changes sign near 1.019679; GCC15/block64 crosses against both controls before 2. GCC10's Hcub/partial losses reverse below their respective roots within the region. These fixed-event boundaries neither identify real RAM prices nor establish physical robustness.

VI. COMPILER ACCOUNT AND INTERVENTION

A. Admitted account of the original block1 binaries

Corrected direct itself is compiler-sensitive. GCC15 executes 242,490 instructions, 43,182 RAM reads and 36,500 RAM writes; GCC10 executes 198,670, 29,304 and 17,225. Both retain two native partials and a final wide reduction, eleven native MULs/output, nine saved/restored ABI words/call, and four stores/reloads at the partial-to-wide interface. The added traffic therefore does not arise from extra MULs, preserved registers or wide per-tap arithmetic.

GCC15 preloads history and maintains a sliding sample window in registers and stack slots. Its common tail rotates samples before testing for another output; GCC10 loads mostly

TABLE V

HCUB MINUS ORDINARY TRANSPOSITION, WHOLE 257-OUTPUT STREAM; PERCENTAGE DENOMINATOR IS ORDINARY TRANSPOSITION. VECTOR COORDINATES ARE $(\Delta nonMUL, \Delta MUL, \Delta B, \Delta R, \Delta W, \Delta F)$; NONMUL IS $N_I - N_M$.

Compiler/block	Δv	ΔE, nJ	Relative
GCC10/1	(0, −2570, 0, 9509, 5397, −1028)	+9166.318770	+3.410536%
GCC10/64	(−9324, −2570, 0, 9257, 5145, −1028)	−268.042614	−0.172202%
GCC15/1	(−17990, −2570, −257, 7196, 5140, −1028)	−11452.561472	−4.284890%
GCC15/64	(−17234, −2570, −5, 7196, 4888, −1028)	−10360.343828	−6.686823%

TABLE VI

EXACT COEFFICIENTS, ROUNDED ROOTS; CELL ABBREVIATES COMPILER/BLOCK. THE FOUR NEGATIVE ROOTS LIE OUTSIDE THE PERMITTED $\lambda \geq 0$ DOMAIN. EXACT RATIONAL ROOTS AND ENDPOINT WINNERS ARE IN THE MACHINE-READABLE ANALYSIS.

Pair (left − right)	Cell	A, nJ	B_{RAM}, nJ	Root (display)	Strict lower on [0.5,2]?
Partial − wide	10/1	−87520.461808	−42645.678958	−2.052270	Yes
Partial − wide	10/64	−64652.689002	−41894.635425	−1.543221	Yes
Partial − wide	15/1	−62253.689535	−40885.154171	−1.522648	Yes
Partial − wide	15/64	−69714.748911	−40667.649816	−1.714256	Yes
Hcub − ordinary	10/1	−4898.171738	14064.490508	0.348265	No; higher throughout
Hcub − ordinary	10/64	−13888.913330	13620.870716	1.019679	No; changes sign
Hcub − ordinary	15/1	−22782.813560	11330.252088	2.010795	Yes
Hcub − ordinary	15/64	−21526.654040	11166.310212	1.927822	No; changes sign
Hcub − partial	10/1	−32436.319000	43973.275423	0.737637	No; changes sign
Hcub − partial	10/64	−16661.004146	24064.520307	0.692347	No; changes sign
Hcub − partial	15/1	−94335.682384	13297.172804	7.094417	Yes
Hcub − partial	15/64	−16351.634074	8374.315001	1.952594	No; changes sign

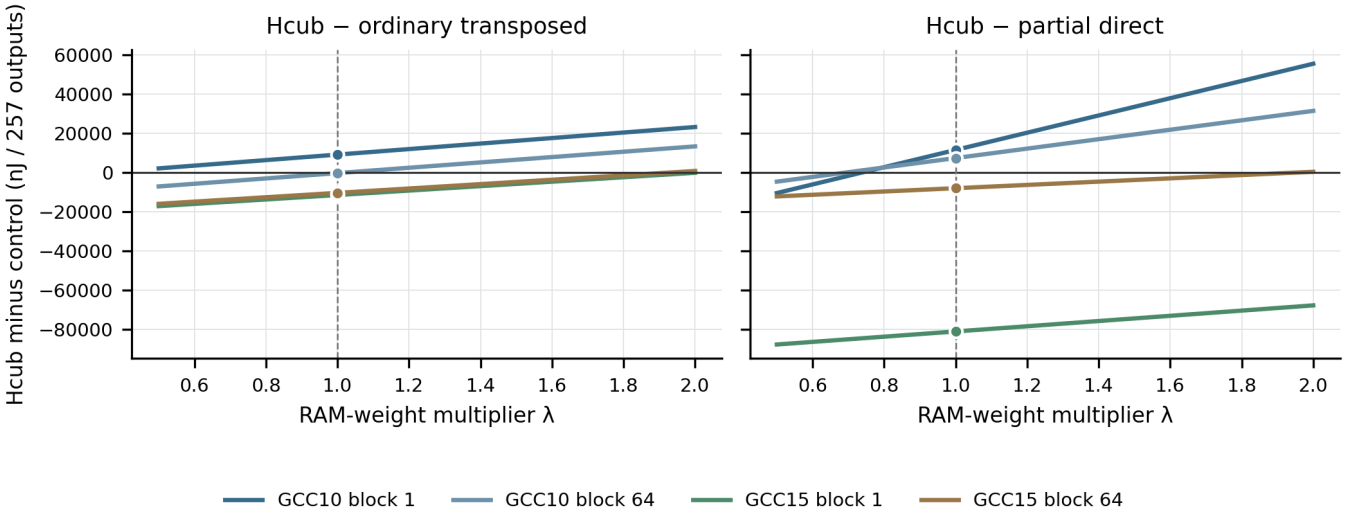


Fig. 2. Saved-event RAM-weight sensitivity for Hcub minus ordinary transposition and Hcub minus corrected direct. Negative differences favor Hcub. Lines reuse fixed 257-output event vectors; lambda scales both RAM weights on one arbitrary, uncalibrated path. Zero crossings are algebraic ranking boundaries, not physical uncertainty bounds.

near use and carries one sample forward. The runtime-block callee remains general despite a one-output caller. For example, `[sp,#40]` holds history sample 45, supplies coefficient 12990 and is replaced by sample 46 for a possible next iteration. Recorded definition/use/update PCs support this value chain. Stack residence alone does not establish a register-allocation spill.

Aggregate logs are not per-PC traces. Executed SP-relative counts follow from checked paths: all 257 calls use $b = 1$, copy-in/convolution each iterate once, 47-step history copy and alternate quantizer paths have no direct-SP access, and quantizer paths rejoin before the common tail. SP stays fixed; no internal calls occur; GCC10's zero-block store is unreachable. Thus GCC10 executes 9/9 direct-SP loads/stores per call

versus GCC15's 63/84: exactly 13,878 additional reads and 19,275 writes, reconciling all added RAM traffic. Convolution, input/output, history and ABI work reconcile the remaining RAM events.

Each extra transfer adds one instruction and one RAM event. Their 59,909.909673 nJ accounts for 85.0151% of the compiler increase; MUL, Flash, Flash-write and BL deltas are zero. Independent checks cover two rows/29 hashes, with manual path/value review providing semantic acceptance. The residual instruction/branch cost stays explicit. This additive account predicts neither a compiler pass nor the counterfactual saving from deleting transfers.

B. Fixed-block specialization contracts and reverses the compiler gap

The two general controls reproduce the historical loadable binaries and all ten saved counter fields exactly, admitting the historical comparison. All four performance and four separate 361-sample signed-edge identities pass output/history/guard and raw-evidence admission. Specialization lowers modeled cost by 23,593.136102 nJ (8.856479%) for GCC10 and 105,920.279148 nJ (31.443058%) for GCC15, each relative to its own general control.

The compiler gap changes from $G_g = 70469.699925$ nJ to $G_s = -11857.443121$ nJ, a contraction of 82,327.143046 nJ. GCC15 therefore becomes the less costly specialized direct build in this pair. Both strict numeric predicates hold: lower GCC15 specialized cost and $G_s < G_g$.

Independent emitted-code review finds the identified next-output preparation removed in both specializations. GCC15's clustered history preloads, stack-resident sliding window and sample-rotation tail disappear; its specialization loads history near use and proceeds from output storage to the required 47-word history shift. GCC10's smaller carried-sample recurrence and output-loop pointer/test also disappear, while the sample needed by the first history shift remains. Thus removing next-output preparation does not remove required retained-history movement.

Other changes accompany fixed-bound knowledge: run-time copy-in and multiple-output loop control, high-register saves/restores, stack locals and stack-materialized accumulator pairs disappear, while arithmetic scheduling and quantizer branches also change. The manual preparation-removal ruling and both numeric predicates support the stronger conjunction for this fixed pair.

The total differences belong to this block-knowledge intervention. The old 85.0151% additive fraction does not partition the new savings, and smaller frames alone do not prove eliminated traffic. No other-form ranking, compiler-pass causation or physical saving follows from these direct pairs.

VII. Prospective synthetic boundary results

A. Wide-required reversal and unavailable native cells

Strengthening direct arithmetic reverses the old-control Hcub advantage in 24 of the 32 complete widening-required fixed-compiler cells: all 16 GCC10 cells and the eight GCC15/block64 cells. Both fresh wide designs supply witnesses under both amplitudes and both measured regimes. The eight GCC15/block1 cells retain Hcub's advantage over the stronger nontransposed set. Thus the frozen existential wide-design extension is observed, rather than inferred from the earlier development filter. The complete panel remains **incomplete**: eight native-fit fast-CMSIS/block64 identities are unavailable because pointer-object aliasing remains unresolved. Those cells contribute neither a win nor a loss to an old/new-control comparison.

The frozen 288-identity inventory contains 280 admitted executions, eight unavailable and zero failed identities. All 280 binaries have distinct hashes; no binary deduplication occurred. The separate 72-identity correctness-only inventory admits 70 and leaves the same native fast-CMSIS/block64 form unavailable under each compiler. Source proofs, edge/reference/state/guard checks and the independent saved-evidence audit support admission. No missing condition was replaced by another coefficient array, implementation or flag.

For example, Wide2800/GCC10/block64/ADC/post-startup gives 661.305273 nJ/output for Hcub, 904.008955 for D_{old} and 515.467942 for D_{new}. The new direct control changes the old-control implementation decision. Ordinary transposition can still win a cell where that reversal occurs: the predicate asks whether strengthening the nontransposed comparator changes Hcub's status, not whether direct is always the best realization. Figure 3 displays both margins for every wide cell.

Among the 40 complete fixed cells, the disjoint all-form outcomes are ordinary transposition in 16, first direct alone in eight, balanced direct alone in four, Hcub in four, and exact first/balanced ties in eight. Counting tied membership instead would give 16 first-direct and 12 balanced-direct appearances; those counts overlap and must not be summed as distinct winning cells.

When each form may select its compiler before comparison, only 20 of 24 cells have known old/new and all-form minima; the four native/block64 cells remain unknown. The wide-design reversal occurs in eight of 16 complete selectable cells, all at block64. Ordinary transposition wins all eight wide/block1 cells; first and balanced direct tie in all eight wide/block64 cells. Hcub wins the four complete native/block1 cells. These selectable decisions answer a different question from the 24/32 fixed-compiler result.

B. Narrowing benefits do not imply incremental-sharing benefits

Every admitted certified direct comparison improves on its wide counterpart, including the native-fit design. The first/balanced partition choice can change which direct implementation wins: it ties under GCC10 for Wide2800 and under GCC10 for both wide block64 groups, favors first under GCC15 for Wide2800 and favors balanced under GCC15 for Wide4600. No claim of optimal partitioning follows.

In the fresh wide designs, Hcub costs more than ordinary transposition in **all 32** cells, by 0.117310% to 7.657365%.

TABLE VII
ACCEPTED ADDITIVE ACCOUNT OF THE TWO ORIGINAL DIRECT BINARIES.

GCC15 − GCC10, block1	Total	Extra direct-SP transfers	Residual
Instructions	43820	33153	10667
RAM reads	13878	13878	0
RAM writes	19275	19275	0
Taken branches	131	0	131
Estimated cycles	77235	66306	10929
Modeled energy, nJ	70469.699925	59909.909673	10559.790252

TABLE VIII
THE FOUR FIXED INTERVENTION ROWS, 257 MEASURED OUTPUTS EACH. CYCLES ARE THE EVENT-DERIVED ESTIMATE. PROCESS-SYMBOL SIZES INCLUDE LITERAL POOLS; PROCESS FRAMES, CONSERVATIVE LIVE CALL-CHAIN BYTES AND LINKED FLASH-IMAGE BYTES ARE DISTINCT RESOURCE MEASURES. STATE ALLOCATION STAYS 98 BYTES, STATIC RAM ENDS AT `0x20000488`, AND INITIAL SP IS `0x20001FF0` IN EVERY ROW. THE SEPARATE EDGE WORKLOAD SUPPLIES CORRECTNESS EVIDENCE ONLY.

Compiler/form	Total energy, nJ	Energy/output, nJ	Cycles/output	Symbol, B	Frame, B	Stack chain, B	Flash image, B
GCC10 general	266394.093850	1036.552894	1065.108949	1036	72	144	2166
GCC10 specialized	242800.957748	944.750808	968.108949	888	20	92	2018
GCC15 general	336863.793775	1310.754061	1365.634241	1372	240	312	2490
GCC15 specialized	230943.514627	898.612897	920.105058	884	16	88	2002

TABLE IX
COMPLETE COST-COORDINATE AND CYCLE REDUCTIONS OVER THE 257-OUTPUT STREAM. VECTOR COORDINATES ARE $(\Delta nonMUL, \Delta MUL, \Delta B, \Delta R, \Delta W, \Delta F)$. INSTRUCTION DELTAS EQUAL NONMUL DELTAS; FLASH-FETCH DELTAS ARE 17,219 AND 72,605. BL AND FLASH-WRITE DELTAS ARE ZERO. DIAGNOSTIC RAW TICK DELTAS FALL BY 23,901 AND 112,828, RESPECTIVELY; THEY ARE NOT THE CYCLE ESTIMATE.

General − specialized	Δv	Estimated cycles	Modeled energy, nJ
GCC10	(17219, 0, 257, 3855, 3341, 0)	24929	23593.136102
GCC15	(72605, 0, 645, 17733, 22873, 0)	114501	105920.279148

TABLE X
EVERY FIXED DESIGN/COMPILER/BLOCK GROUP, EACH COVERING ADC/FULL × STARTUP/POST-STARTUP. MINIMA AND REVERSAL STATUS ARE INVARIANT ACROSS THOSE FOUR CONDITIONS. THE REVERSAL USES D_{old} AND D_{new}, NOT THE ALL-FORM MINIMUM; ORDINARY TRANSPOSITION IS EXCLUDED FROM BOTH NONTRANSPOSED SETS. NATIVE, WIDE2800 AND WIDE4600 ABBREVIATE THE FROZEN DESIGN IDENTIFIERS.

Design	Compiler/block	Reversals / four cells	All-form minimum
Native	GCC10/1	Not a wide-design test	First direct
Native	GCC10/64	Unknown	Unknown
Native	GCC15/1	Not a wide-design test	Hcub
Native	GCC15/64	Unknown	Unknown
Wide2800	GCC10/1	4/4	Ordinary transposed
Wide2800	GCC10/64	4/4	First = balanced direct
Wide2800	GCC15/1	0/4	Ordinary transposed
Wide2800	GCC15/64	4/4	First direct
Wide4600	GCC10/1	4/4	Ordinary transposed
Wide4600	GCC10/64	4/4	First = balanced direct
Wide4600	GCC15/1	0/4	Ordinary transposed
Wide4600	GCC15/64	4/4	Balanced direct

Thus a reversal against old direct controls does not demonstrate beneficial incremental sharing. The native control retains the original compiler split for this contrast: Hcub is more costly with GCC10 and less costly with GCC15. The original GCC15-wide Hcub advantage over ordinary transposition therefore does not generalize to these two wide arrays. The exact event vectors, every direct correction and all per-cell costs remain in the panel results and summary.

C. Workload boundaries and admitted resources

Block64 lowers cost/output in all 136 known within-form block pairs, by 31.554538% to 55.482311%; eight pairs are unknown. Full-range input is equal in cost to its paired ADC input in 28 of 140 known amplitude pairs and lower in 112, with changes from −0.046876% to zero; four are unknown. Post-startup cost/output is lower in 126 of 140 known regime pairs and higher in 14, spanning −0.375190% to +0.025079%; four are unknown. These exact deterministic pairs are work-

TABLE XI

MIN–MAX OVER EACH GROUP'S FOUR AMPLITUDE/REGIME CELLS, ROUNDED ONLY FOR DISPLAY. EACH PERCENTAGE IS $100(E_{left}/E_{named} - 1)$; NEGATIVE FAVORS THE LEFT. A REPEATED ROUNDED ENDPOINT CAN HIDE A NONZERO RANGE. DIRECT AND ORDINARY/HCUB PAIRS REMAIN KNOWN IN NATIVE/BLOCK64 DESPITE UNAVAILABLE CMSIS.

Design	Compiler/block	First / wide, %	Balanced / wide, %	Hcub / ordinary, %
Native	GCC10/1	−40.550 to −40.548	—	+0.448 to +0.448
Native	GCC10/64	−58.108 to −58.022	—	+0.791 to +0.794
Native	GCC15/1	−27.626 to −27.625	—	−1.454 to −1.454
Native	GCC15/64	−45.856 to −45.763	—	−1.987 to −1.983
Wide2800	GCC10/1	−34.339 to −34.321	−34.339 to −34.321	+7.656 to +7.657
Wide2800	GCC10/64	−54.969 to −54.882	−54.969 to −54.882	+7.220 to +7.222
Wide2800	GCC15/1	−24.616 to −24.586	−24.517 to −24.487	+0.117 to +0.117
Wide2800	GCC15/64	−43.107 to −43.020	−42.691 to −42.605	+0.129 to +0.129
Wide4600	GCC10/1	−29.294 to −29.277	−29.357 to −29.340	+4.512 to +4.513
Wide4600	GCC10/64	−45.743 to −45.653	−45.743 to −45.653	+3.368 to +3.374
Wide4600	GCC15/1	−23.741 to −23.727	−23.964 to −23.950	+0.571 to +0.572
Wide4600	GCC15/64	−42.254 to −42.167	−42.734 to −42.646	+1.377 to +1.382

load contrasts, not independent statistical samples. Neither amplitude nor measured regime changes the known minima or reversal statuses in the grouped table. The small cost changes still deserve reporting rather than an assumed monotonic startup benefit.

The source quantizer reaches neither clamp threshold on ADC inputs or the native full-range input. Wide2800/full reaches the positive/negative thresholds 6/6 times in startup and 31/22 post-startup; Wide4600/full reaches them 12/14 and 55/53 times, respectively. Counts apply only to the measured 257 or 1025 outputs, derive from exact source sums and do not identify executed target branch frequencies. Saturation coverage thus changes while the full-int16 arithmetic contract and the finite implementation decisions stay fixed.

All admitted executables respect the original memory limits. The maximum static RAM end is `0x20001210`, below `0x200017f0`; the largest conservative live stack chain is 688 bytes within the reserved 2048, with initial SP `0x20001ff0`. The maximum Flash end is `0x0800135e`. These linked-image and call-chain checks strengthen admission for the long fixtures; individual state, process-frame and selected-symbol metrics remain separate diagnostics.

VIII. DISCUSSION AND LIMITS

A. Contribution and pursuit verdict

Arithmetic-fair direct correction changes two original implementation decisions; the fresh wide-required panel supplies 24 complete fixed-cell witnesses to the same comparator boundary. Yet incremental sharing loses in every fresh wide cell. That adverse result matters: a nominal win over old direct/library controls can coexist with no gain from sharing within transposition. The compiler split, tied partitions and unavailable native cells prevent a simple width-based selector.

The planned upgrades have distinct contribution roles. U1's controlled framing and U4's portable evidence are supporting scholarship. U2 adds exact artifact-specific event tradeoffs and sensitivity boundaries, with no new analysis method. U3 supplies bounded prospective wide-design witnesses while leaving the full panel incomplete; it does not establish representative prevalence or beneficial sharing on fresh wide designs.

U5 passes its stronger local conjunction: reviewed preparation disappears, GCC15 cost falls and the compiler gap contracts past zero. This strengthens the account that callee-visible block knowledge matters, while broader code changes prevent exclusive attribution to rotation removal. The combined outcomes support pursuing the work as a bounded empirical case study with auditable controls and mechanism evidence. They establish neither a new algorithm nor field priority, an implementation selector or publication acceptance.

B. Internal and external validity

Arithmetic proofs, graph algebra, UBSan, finite output/state/guard checks, binary identity and emitted-code review address different failure modes. Output agreement alone proves neither C-language safety nor model validity. The tested direct cuts/schedules/options are finite controls, not an exhaustive optimization search. Their pre-cost freeze prevents outcome-driven replacement.

The original ADC stream spans −1024 to 1023: its bound $1024 \times 72509 = 74249216$ limits output magnitude to 2266 without saturation. The new full-range inputs exercise source clamps but are still finite synthetic stimuli. Three gain-normalized arrays at one length cannot establish application prevalence. Neither static full-domain legality nor a post-startup segment turns those stimuli into exhaustive performance coverage or proven steady state. The chronology separates old development evidence from the fresh panel.

Single-cycle MUL, zero waits and frequency/prefetch settings are model assumptions. Board current, peripheral energy and physical uncertainty are unmeasured. Reweighting fixed events does not simulate another configuration, which could change the events themselves. The block1 direct intervention has a satisfied $b = 1$ precondition and cannot establish block64 behavior, other-form rankings or compiler-pass causation.

Resources constrain adoption separately from modeled energy. Original direct state allocation is 98/224 bytes at blocks 1/64, versus transposed192. Its process frames and selected symbols (Appendix A) exclude full call-chain/image costs.

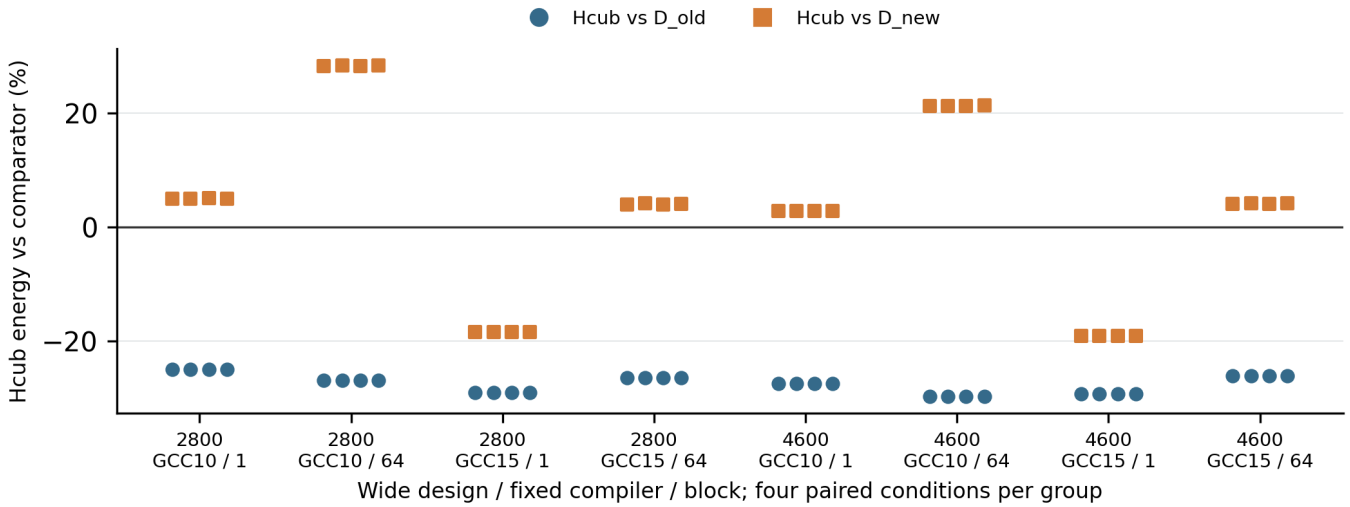


Fig. 3. Every complete wide-design fixed-compiler boundary. Hcub relative to the old and strengthened nontransposed minima crosses from below to above zero in 24 of 32 cells. GCC15/block1 supplies the eight nonreversals. Within each group, points run left to right as ADC/startup, ADC/post-startup, full/startup and full/post-startup. Eight unavailable native fast-CMSIS/block64 identities leave the full panel incomplete.

The new panel additionally admits actual linked-memory and conservative stack-chain bounds. These metrics remain distinct rather than summed as RAM.

C. Evidence portability and reproduction levels

The original 531-file raw-evidence slice includes source/binaries/disassembly/ logs, certificates, Hcub notices, checker code and dependency identities. A restored clean copy regenerated the numerical report and checked 24 historical, four partial-direct and two explanation rows, including 29 explanation hashes, with zero skips. Explicit copied roots prevent original-workspace fallback. Those check levels overlap in underlying binaries; they are not additional disjoint target experiments. The extended package was exercised separately.

Table regeneration, saved-binary verification and new target execution are distinct levels. Executable toolchain trees are excluded; rebuilding needs separately acquired pinned GCC10/GCC15, simulator and CMSIS/Hcub dependencies, with acquisition/identity instructions. Hash consistency detects changed bytes, not authenticity or physical validity. No public archive or DOI is claimed.

The upgraded package contains 7,266 manifested payload files. A fresh relocated copy passed all seven verification commands: original numerical regeneration, 24 historical/four partial-direct raw checks, two explanation checks with 29 evidence hashes, exact U2 regeneration, panel performance and correctness checks, and the intervention audit. The panel audits admitted 280 performance identities with eight scientifically unavailable and zero failed, and 70 correctness identities with two expected unavailable and zero failed. The intervention audit admitted four performance and four correctness rows. All seven commands exited successfully, with zero integration skips and zero target-tool executions. Scientific unavailability remains an explicit result of the frozen panel and does not represent skipped verification. This exercise establishes portability of the saved evidence and regenerable analyses at the stated levels, rather than a fresh target rebuild or physical replication.

IX. Conclusion

Arithmetic-fair direct correction overturns two original Hcub decisions and supplies 24/32 complete-wide prospective reversal witnesses, with unavailable native cells keeping the full panel incomplete. Beneficial incremental sharing does not extend to either fresh wide design: ordinary transposition wins that contrast throughout. Saved events expose RAM-sensitive margins, and a matched literal-one direct specialization removes next-output preparation, lowers cost and reverses the compiler gap. These finite results make comparator arithmetic, state representation and visible API knowledge necessary parts of interpreting this artifact's nominal winners. They support an auditable model case study, without a new algorithm, physical saving or general implementation rule.

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TABLE XII
EVERY ORIGINAL PRINCIPAL-ROW RESOURCE DIAGNOSTIC.

Cell	Form	Cycles/out.	Code, B	State, B	Frame, B
GCC10/1	CMSIS	1585.11	1494	112	8
GCC10/1	Wide direct	1606.11	1736	98	440
GCC10/1	Partial direct	1065.11	1036	98	72
GCC10/1	Ordinary transposed	1106.11	1540	192	272
GCC10/1	Hcub	1150.11	1504	192	344
GCC10/64	CMSIS	924.32	1494	236	8
GCC10/64	Wide direct	1029.33	1720	224	440
GCC10/64	Partial direct	583.31	1196	224	72
GCC10/64	Ordinary transposed	630.64	1540	192	272
GCC10/64	Hcub	636.40	1504	192	344
GCC15/1	CMSIS	1550.11	1450	112	8
GCC15/1	Wide direct	1798.11	1872	98	608
GCC15/1	Partial direct	1365.63	1372	98	240
GCC15/1	Ordinary transposed	1100.11	1532	192	272
GCC15/1	Hcub	1062.11	1352	192	344
GCC15/64	CMSIS	851.98	1450	236	8
GCC15/64	Wide direct	1074.65	1884	224	608
GCC15/64	Partial direct	612.39	1312	224	232
GCC15/64	Ordinary transposed	627.57	1532	192	272
GCC15/64	Hcub	593.49	1352	192	344

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APPENDIX A
ORIGINAL PRINCIPAL-ROW RESOURCE DIAGNOSTICS

Cycles are estimated per requested output. Code is selected kernel/coefficient symbols, excluding fixture and possibly helper code; frame is the compiler's static `probe_process` report; state is allocated state including block capacity. These are not total image, peak stack or SRAM high-water measurements.

APPENDIX B
EDITABLE EVIDENCE MAP

The original manuscript/results remain preserved under `research/fir-paper/`. Its source manifest records six frozen numerical inputs. The original coefficient and timed-input SHA256 values are `63796e3e365dbdee70192acf68f16f403802d961df48ae814ffb23cca411fb70` and `d952c2754c9ae922243966a9c2533a848b345aa2d2d6cb5b4e68d9df1615f310`. The saved-event analysis retains source digests, 12 oriented identities, exact difference vectors, cost components, cycle and separate resource deltas, rational boundaries and endpoint winners. The portable-evidence instructions, evidence manifest and clean-copy report identify the package and verification levels.

The frozen panel protocol, coefficient design, inputs, certificates, graph manifest and source manifest bind the prospective preparation. The panel identity inventory, exact summary and independent raw audit preserve every condition, event vector, tie, unavailable identity and paired contrast. These panel results have passed independent scientific review. The intervention inventory binds four performance and four correctness rows to their raw artifacts. Its measurements are admitted, and the independent emitted-code review accepts the preparation-removal ruling and stronger conjunction. The automated measurement record retains a pending-review field distinct from that manual ruling. The clean-copy report records the extended package verification described in Section 8.3. The historical 12/24 filter-bank statistic belongs to its original candidate set; no revised bank-wide result is inferred here.